Managers see OSA fall, but not why

~2 days
to work out why a single OSA decline happened

~40%
of high-impact issues fixed within 14 days today

6 feeds
OSA, POS, store stock, DC stock, audits and promos, each in its own system

WHAT DISCOVERY SHOWED

Same chart, different causes
A DC shortage, a store ordering miss and a shelf execution gap look identical on an OSA report.

Fragmented data, manual stitching
Every investigation means pulling reports into spreadsheets and waiting on analysts and supply chain.

Gut-feel prioritisation
Stores are picked by judgment, with no shared view of the sales each issue is losing.

No feedback loop
Nobody records which fix was tried or whether OSA recovered, so the same issues repeat.

Source: Step 1 discovery (persona, empathy map, 5 Whys). The 2-day and 40% baselines are assumptions to validate in the pilot.

We chose diagnosis ranked by lost sales

JOB TO BE DONE

"When I see OSA drop, help me find which store-SKUs matter most and why they dropped, so I can trigger the right fix with the right owner this week."

Why AI, and where

Signals conflict and causes combine, so diagnosis needs judgment that fixed rules handle poorly. The model only weighs evidence; code computes every number.

V1 SCOPE, EACH MAPPED TO A CAPABILITY

Opportunity	AI capability used
Flag at-risk store-SKUs	Rules and statistics, no LLM
Rank by estimated lost sales	Deterministic calculation
Diagnose the root cause	LLM reasoning over an evidence pack
Recommend action and owner	Retrieval from the approved playbook
Plain-language Q&A	Read-only tool calling
Draft briefs and escalations	Templates; the manager sends

Chosen on the impact × feasibility opportunity map: these six fix the slow, unprioritised diagnosis loop first.

Deferred: OOS prediction, shelf-photo recognition, action-effectiveness tracking, SKU/store code matching, weekly executive summary.

From weekly refresh to approved action

1
Check feeds
Gaps set aside, never read as out-of-stock

2
Flag and rank
By estimated lost sales, inputs shown

3
Evidence pack
Signals plus one rule check per cause

4 · AI
Diagnose
1–3 causes from a fixed list, cited

5
Validate, gate
Unsure or unproven: “Needs investigation”

6
Manager decides
Playbook action, draft the manager sends

WALKTHROUGH · #1 ON THIS WEEK’S WORKLIST

Flag
Store A113, Cetirizine: OSA 40.4% vs 96.6% baseline; est. lost sales $1,590.72

Diagnosis
Delisting, high confidence, citing “not ranged on planogram” and “0 facings”

Action
PB-06: Key Account Manager confirms ranging with the buyer within 10 days

Decision
Manager accepts, edits the draft brief and sends it; all logged

Prototype: Python, Groq gpt-oss-120b, Streamlit · synthetic data, 62 stores × 20 SKUs × 12 weeks · diagnosis in under 3 seconds.

OSA ROOT-CAUSE COPILOT · 4 · SUCCESS METRICS

Success means issues fixed in 14 days

NORTH STAR

40% → 70%

of high-impact OSA issues resolved within 14 days in the pilot. Resolved = back at target for 2 weeks in a row.

Layer	Metric	Target
Output	Flag to diagnosed cause	Under 30 min (from ~2 days)
Trust	Diagnoses accepted without edit	At least 70%
Guardrail	Numbers not found in the data	Zero, every week
Guardrail	"Needs investigation" rate	5% to 20%
Business	OSA vs control regions	+2 pts after 8 weeks

ALPHA LAUNCH EVAL · 60 SEEDED SCENARIOS · EVERY THRESHOLD MET

100%
top-1 root cause

165/165
citations grounded

0
made-up numbers

20/20
off-topic declined

2.7 s
p95 diagnosis

Eval ran on synthetic data with clean patterns; the 40% baseline is assumed. Real-data accuracy is proven in the shadow-mode pilot.

Guardrails make errors rare and visible

Risk	Guardrail built in
Confident wrong cause	Fixed cause list, 2+ cited signals, confidence gate, approval on every item
Made-up or altered numbers	All numbers computed in code; the validator blocks any value not in the data
Data gap read as out-of-stock	Freshness checks run first; gaps are excluded and the data owner notified
Action sent without approval	No send or write capability: drafts only, and the manager sends them
Cross-retailer data leak	Scoped to one retailer at load, before the model sees anything
Blame or prompt injection	People's names and free-text notes are never loaded into the evidence

Kill switch

Roll back at once if a shown number mismatches the data, anyone sees another retailer's data, anything leaves without approval, or acceptance stays under 40% for 2 weeks.

Next: shadow-mode pilot

1 retailer, 1 category, 3–5 managers for 4 weekly cycles; nothing is sent from the tool. Gate: top cause agrees with managers 70%+ of the time, zero numeric errors.

Open before any real-data pilot: legal confirmation that retailer data-sharing terms allow an external LLM API.